

Course 101 Mini Project: Detecting the anomalous activity of a ship's engine

Instructions: Use the [solution Colab Notebook](#) and 'Details' column to decide one whether the criterion was met

Skills	Criteria (markers)	Criterion has been met	Feedback
Initial data exploration and assessment	Missing and duplicate values in the data are identified or reported as non-existent.	☒	Keep doing: Good use of comments in code. Good use of headings in code for better structure. Areas of improvement:
	The descriptive statistics of the data set are generated.	☒	
	The mean and medians for each feature are documented providing insights into the central tendency of the data.	☒	
	The range values beyond the 95th percentile for at least two features are identified.	☒	
	Visualisations are created to facilitate understanding of the data.	☒	
	The Interquartile Range (IQR) method was used to detect outliers in the data.	☒	
Anomaly detection: Statistical methods	A new binary column was created for each feature, indicating whether values are outliers based on IQR calculations.	☒	
	Identify the number of features that must simultaneously be in outlier condition, in order for a sample to be classified as an outlier, such that the total percentage of samples identified as outliers falls within the 1-5% range.	☒	
	Thoughts and observations about the IQR method are recorded to understand its effectiveness in anomaly detection.	☒	
Anomaly detection: Machine learning (ML) models	Features in the dataset are scaled to improve model performance.	☒	Areas of improvement: Use unscaled data as an input to IF.
	One-class SVM was used for outlier detection to explore various approaches.	☒	
	Model settings for One-class SVM were adjusted to obtain the expected range of outliers.	☐	
	Isolation Forest was used for outlier detection to explore various approaches.	☒	
	Model settings for Isolation Forest were adjusted to obtain the expected range of outliers.	☐	
	Thoughts and observations about One-class SVM and Isolation Forest methods were recorded to understand its effectiveness in anomaly detection.	☒	
Report: Communicating business impact and insights	The report is between 800-1000 words.	☐	Areas of improvement: In the introduction include that the aim is to identify 1-5% outliers. In the methods of IQR explain what is the method used to identify number of features that must be simultaneously be in outlier condition in order for a sample to be considered as an outlier. Present results of models with different parameters as tables rather than text for better presentation of results, for example, parameter 1, parameter 2, number of outliers, percentage of outliers. We cannot perform Feature importance as we are operating in an unsupervised machine learning scenario.
	The report documents the approach used.	☒	
	Major inferences regarding the data analysis are evident.	☒	
	Conclusions drawn from the analysis are supported by the data.	☒	
	Visualisations are created to effectively convey the story of the data.	☒	
	The effectiveness of different methods is assessed based on gathered evidence.	☒	
	The effectiveness of 2D visualisations in highlighting outliers is evaluated.	☒	
	The report is clear, well-organised, and engaging to facilitate learning from the analysis.	☒	
			Overall Feedback:
			Learner performed majority of analysis requested and showed good understanding of concepts. Areas of improvement are highlighted in the sections above.